Flow Map of Internal Migration Patterns in Argentina, 2010

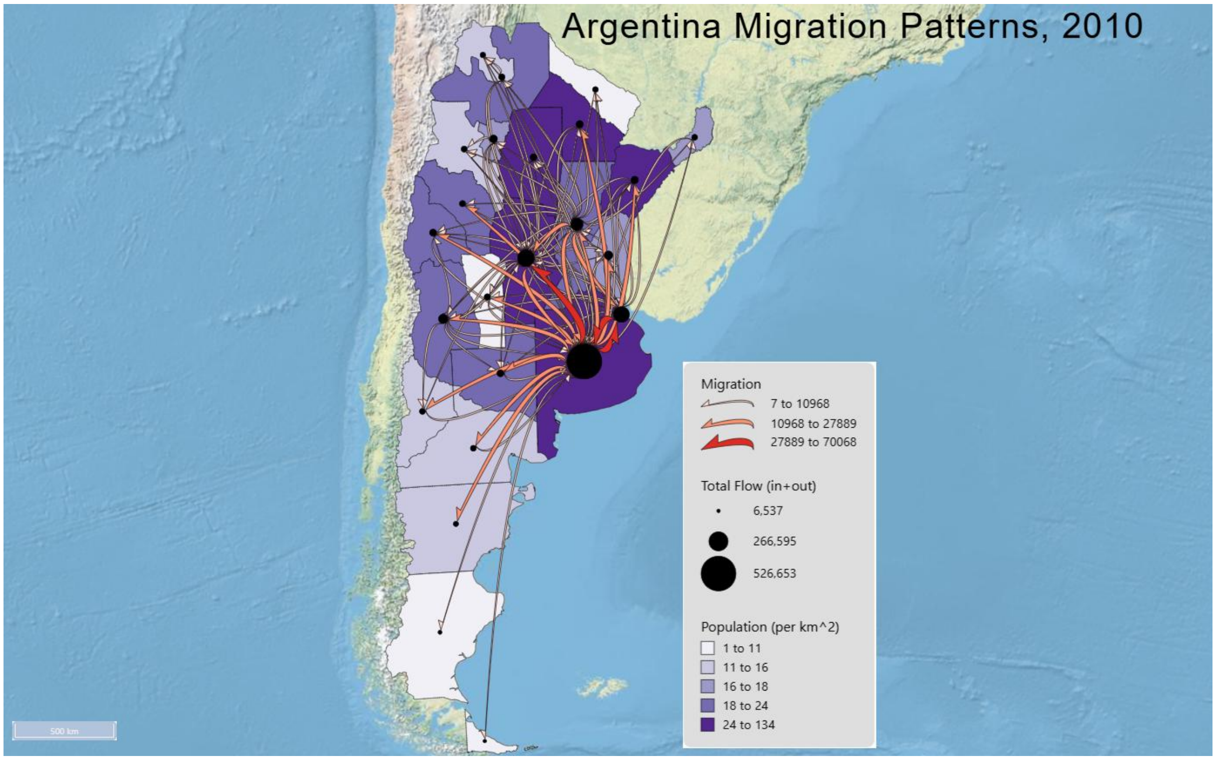

Figure 1 Flow map of internal migration patterns in Argentina in 2010, showing major origin–destination migration flows, province-level population density, and total connected flow at each node.

Justification of Flow Map Design

This flow map was designed to emphasize the dominant internal migration structure of Argentina while still preserving regional context. FlowMapper was developed to support this kind of cartographic workflow by allowing the customization of flow lines, node symbols, choropleth layers, base maps, and map elements so that the final map can be tailored to the reading task rather than relying on one default symbolization.

For the flow layer, I used the Curve Half Arrow style. This choice is appropriate for a directed origin–destination migration map because the half-arrow explicitly communicates direction, while curved flow lines help separate connections that would be harder to follow if they were drawn as straight segments. FlowMapper's curved line options are intended to improve the readability of direction and connection structure, and the platform notes that curved paths with varying thickness can reduce overlap at origins and destinations. In addition, research summarized by Koylu et al. 2017, 2022 indicates that curved flows are generally easier to interpret than straight lines in dense flow maps, and that asymmetric curvature can strengthen the perception of direction in two-way flow maps.

I displayed only the top 100 flows rather than all flows. This was an intentional clutter-reduction decision. The reference article explains that dense flow datasets quickly become unreadable when every connection is shown, and the assignment instructions specifically emphasize limiting overlap through the “show top flows” setting. Using the top-ranked flows preserves the overall migration structure while removing weaker links that would obscure the dominant pattern.

For flow magnitude, I used Natural Breaks with 3 classes, a red color ramp, and widths ranging from 4 to 16 with a dark stroke. This combination creates a clear visual hierarchy. The article notes that flow maps often contain a few very large flows and many smaller ones, which makes fully proportional line thickness harder to compare visually. It also states that classified line widths are useful when the goal is to improve magnitude comparison. Natural breaks is therefore suitable here because the migration volumes are highly uneven, and three classes keep the legend compact and easy to read rather than over-segmenting the data. The red ramp makes the strongest flows immediately visible, so the main analytical task is to identify major migration corridors rather than minor links. The dark stroke helps separate lines where several flows cross or run close together.

For the node layer, I mapped Total Connected flow (incoming + outgoing) using proportional black circles with a minimum radius of 2 and maximum radius of 20. This follows the logic described in the article, where node symbols are used to show a gross flow measure at each location. In this map, total connected flow is the most useful node attribute because it shows which provinces function as the strongest overall migration hubs, regardless of whether they are net senders or net receivers. A proportional scale is appropriate because the variable is continuous, and the black circles provide a neutral but strong anchor on top of the choropleth and physical base map.

For the choropleth background, I mapped population density using 5 quantile classes with a purple sequential scheme and thin black outlines. The article explicitly notes that choropleth maps are useful for contextualizing region-to-region flows with locational attributes such as population density. Quantiles work well here because Argentina has a limited number of provinces, and the quantile method distributes regions across classes in a way that avoids having most provinces collapse into only one or two categories. The purple scheme remains visually distinguishable to the red flows, which preserves the flows as the primary theme while still allowing the reader to compare migration structure with regional density.

For the base map and layout, I used the Esri World Physical Map, the Albers Equal Area South America projection, and a clear title. That projection is appropriate here because the map covers a large national extent within the South American continent, and an equal-area regional projection is more suitable than Mercator for a thematic map intended to compare provinces across space. The physical base map provides coastline and terrain context without introducing too many labels or urban features that would compete with the migration symbols. The title clearly identifies the subject, place, and year.

Overall, the symbology was designed with visual hierarchy in mind. The flows are the main message, so they are the warmest and most visually dominant symbols. The nodes are secondary anchors that show hub importance, and the choropleth provides regional context in a cooler and more subdued palette. This differs from the example maps in the paper because I used a physical basemap, a red flow scheme, black proportional nodes, and a purple density choropleth instead of reproducing the article’s published color combinations.

Interpretation of Flow Patterns

The map shows a strongly centralized migration structure, with the most intense internal migration focused in Buenos Aires Province and the broader east-central corridor of Argentina. The largest node by total connected flow is Buenos Aires with 526,653, far exceeding the rest of the system. The next highest totals are Córdoba with 226,174, Capital Federal with 210,500, Santa Fe with 155,039, and Mendoza with 104,846. This indicates that the migration system is not evenly distributed across the country; instead, a small number of major provinces dominate the network. These larger nodes serve as the principal migration hubs in the national system. In terms of sending locations, Buenos Aires Province is the strongest source of migrants, with 393,706.93 outgoing migrants. Other major sending provinces include Santa Fe with 109,326.74, Córdoba with 109,704.55, Capital Federal with 87,679.31, and Mendoza with 49,770.81. Buenos Aires in particular functions as the dominant circulation hub rather than a purely one-sided destination.

In terms of receiving locations, Buenos Aires Province is also the strongest attractor, with 132,946.44 incoming migrants, followed closely by Capital Federal with 122,821.17 and Córdoba with 116,469.27. This shows that the main migration destinations are concentrated in the same broad east-central region. Córdoba also stands out as a major interior destination, indicating that migration is not only a simple pull toward the capital region but also toward other important regional metropolitan centers.

The general directional pattern is mainly from the interior toward the east-central part of the country, especially toward the Buenos Aires–Capital Federal area. Many of the strongest links extend from northern, northwestern, and central provinces toward that corridor. There are also important exchanges among the central provinces, especially involving Córdoba and Santa Fe, which suggests that the migration system includes both national-scale concentration and regional circulation. By contrast, the far south and much of the west show fewer major flows and smaller node totals, indicating weaker integration into the highest-volume migration corridors.

The map shows a clear relationship between migration intensity and Argentina's broader population structure. Provinces with larger total connected flow are concentrated in the more populated eastern and central parts of the country, while many peripheral provinces have smaller nodes and fewer major connections. This suggests that internal migration is tied to the country's main urban and economic core, although density alone does not explain everything, since Córdoba also stands out as a major hub. Overall, the map indicates a core–periphery pattern in which Buenos Aires Province, Capital Federal, and Córdoba dominate the migration system, and the strongest flows cluster along Argentina's most urbanized corridor.

References

Koylu, C., & Guo, D. (2017). Design and evaluation of line symbolizations for origin–destination flow maps. *Information Visualization, 16*(4), 309–331. https://doi.org/10.1177/1473871616681375

Koylu, C., Tian, G., & Windsor, M. (2022). FlowMapper.org: A web-based framework for designing origin-destination flow maps. *Journal of Maps*. https://doi.org/10.1080/17445647.2021.1996479

Appendix

Table 1 Total Migration by Region

Region Name	NODE_ID	Total Flow
Buenos Aires	6	526653.3721
Catamarca	10	49171.07362
Chaco	22	70591.63814
Chubut	26	36138.57436
Capital Federal	2	210500.478
Cordoba	14	226173.8194
Corrientes	18	70489.75811
Entre Rios	30	91996.81251
Formosa	34	43606.20871
Jujuy	38	40518.11204
La Pampa	42	63276.63401
La Rioja	46	50551.28469
Mendoza	50	104846.2308
Misiones	54	47041.94815
Neuquen	58	47253.89247
Rio Negro	62	46386.57666
Jujuy	66	54237.38993
San Juan	70	66669.93197
San Luis	74	55956.95398
Santa Cruz	78	14348.98328
Santa Fe	82	155038.7837
Santiago del Estero	86	60214.22676
Tierra del Fuego	94	6537.494429
Tucuman	90	77555.4475